VIP 2025 Cyber Security Cisco Packet Tracer Project

Part 2 – Hybrid Secure Network Architecture

1. Updated Network Diagram (Conceptual Description)

- **Amity Router** (Core Router)
- Subinterfaces:
  - G0/0.10 → VLAN 10 (Faculty) → 192.168.10.1/24
  - G0/0.20 → VLAN 20 (Students) → 192.168.20.1/24
  - G0/0.30 → VLAN 30 (Servers) → 192.168.30.1/24
- G0/1 → WAN Link → 200.1.1.1/30
- **Student Switch** → PCs in VLAN 20
- **Faculty Switch** → PCs in VLAN 10
- **Server** → VLAN 30 (192.168.30.x)
- **Home Router**
- G0/2 → WAN Link → 200.1.1.2/30 (connected to Amity Router G0/1)
- G0/0 → Home LAN → 192.168.40.1/24
- **Remote Faculty Laptop** → Home Router LAN → 192.168.40.2/24 (gateway 192.168.40.1)

This provides both campus-based and remote faculty access, while keeping the server protected.

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2. Network Segmentation

- **VLAN 10 – Faculty**: Campus faculty systems (IP range: 192.168.10.0/24).
- **VLAN 20 – Students**: Student PCs (IP range: 192.168.20.0/24).
- **VLAN 30 – Servers**: Internal servers for academic/research use (IP range: 192.168.30.0/24).
- **VLAN 40 – Remote Faculty**: Home faculty devices via Home Router (IP range: 192.168.40.0/24).

Segmentation ensures that users only access resources they are authorized for, while isolating sensitive servers.

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3. Secure Access Tools (Recommendations)

- **Simulated in Packet Tracer**: Static routing across WAN link (200.1.1.0/30).
- **Real-world Recommendation**:
- **VPN (IPSec/SSL)**: To encrypt remote faculty traffic.
- **SASE or Identity-Aware Proxies**: For role-based access enforcement.
- **Split Tunneling**: To allow remote faculty to access internal services securely, while sending general internet traffic outside the tunnel.

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4. Trust Models & Authentication Flows

- **Faculty (Campus/Remote)** → Trusted access to VLAN 10 and VLAN 30.
- **Students** → Restricted to VLAN 20, cannot directly reach VLAN 30 servers.
- **Servers** → Accessible only via Amity Router gateways, not exposed to the internet.
- **Remote Faculty** → Authenticate via Home Router → Amity Router → authorized access to internal networks.

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5. Updated Topology Pathways

- **Campus Faculty**: Faculty Laptop → Faculty Switch → Amity Router → Servers.
- **Students**: Student PCs → Student Switch → Amity Router → limited access.
- **Remote Faculty**: Remote Laptop → Home Router LAN (192.168.40.0/24) → WAN link (200.1.1.0/30) → Amity Router → internal VLANs.

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6. Risk Analysis & Fallback Strategies

- **Risks**:
- If WAN fails, remote faculty lose access.
- Static routing (in PT) provides no encryption (security gap).
- **Fallbacks**:
- On-campus faculty still have direct access to VLAN 10 and 30.
- Real-world deployments should include VPN failover or redundant links.

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7. Conclusion

This design balances **simplicity, security, and scalability**: - Faculty can work from both campus and remote locations. - Students remain isolated with limited privileges. - Internal servers remain protected from direct internet exposure. - Static routing in Packet Tracer simulates secure connectivity, while real-world solutions would use VPN/SASE.

Deliverables Completed: - ✅ Updated network diagram (conceptual description) - ✅VLAN-based segmentation - ✅Routing + remote access pathway - ✅Security recommendations - ✅Risks and fallback strategies